

ALEXANDER KAMPAS

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SOFTWARE ENGINEER EXPERIENCE

Bloomberg ETN FIXNET July 2022 - November 2024

- Developed and maintained Bloomberg **FixNet**, a global order-routing network processing >\$92B/day in trading volume across buy-side and sell-side institutions. Implemented C++ services handling FIX sessions, order routing, and recovery logic
- Designed and implemented a **C++ in-memory securities metadata cache** for trading infrastructure (FIX routing path), reducing lookup latency from 200ms to sub-millisecond through cache-aligned data structures, advanced warmup techniques, and removal & reduction of RPC dependencies.
- Originated cache's warm-up and testing suite, which further reduced average lookup latency by 20%, saved 1 day a week for entire team, and predicted & prevented catastrophic failures
- Designed network **topology for global FixNet nodes** (Tokyo, London, etc), simultaneously formalized an automated datacenter setup workflow, saving weeks of engineering time every quarter
- Migrated trading infrastructure services from Solaris/AIX to Linux, rewriting components in modern C++ and improving operational reliability

Apple iOS System UI Performance February - August 2021

- Optimized memory locality and allocation behavior in iOS System UI processes
- Developed tools for profiling and optimizing iOS and macOS memory management, focusing on allocation behavior, cache miss patterns, and memory footprint

EDUCATION

Carnegie Mellon University Fall 2017 – Spring 2022

- Bachelors of Computer Science and Arts – Computer Science and Music Performance
- University Honors | BCSA College Honors | 3.67 GPA | Full Scholarship
- Relevant Courses | Distributed Systems, Computer Systems, Computer Security, Great Ideas in Theoretical Computer Science, Machine Learning, NLP, Parallel and Sequential Data Structures and Algorithms

PROJECTS

- **Async Multi-Venue Trading Bot** with live market data ingestion, automated execution, risk controls, OMS, paper trading, and a real-time monitoring dashboard. Event-driven architecture for low-latency trading. Designed and implemented several strategies for illiquid markets (Python)
- **Distributed Bitcoin miner** and underlying proprietary server-client oriented network protocol (Go)
- **Proxy server** supporting web object caching, multiple concurrent connections (POSIX, C, sockets)
- **Dynamic heap memory allocator** (alternative to malloc, realloc, calloc, free) (C)
- **Automatic tailor** that generates clothing patterns for perfectly-fitting garments from a person's photos (C++, Python)
- **Algorithm and program that composes music** in the style of J.S. Bach (Python)
- **Music director** for *Le Cabaret du Hot Club d'Alex*, a monthly improvisational jazz concert in Manhattan

SKILLS

- **Languages** | C++, Python, C, Go, x86 Assembly, Standard ML, Obj-C
- **Systems** | GNU/Linux & UNIX, Low-Latency Systems, HPC, multithreading, Distributed Systems, Concurrency (locks, lock-free), NUMA/cache awareness, Systems-level C, Binary and Net Exploit Development, Functional Programming
- **Networking** | TCP/UDP, FIX protocol, kernel bypass (DPDK)
- **Tools** | gdb, perf, standard Linux utils